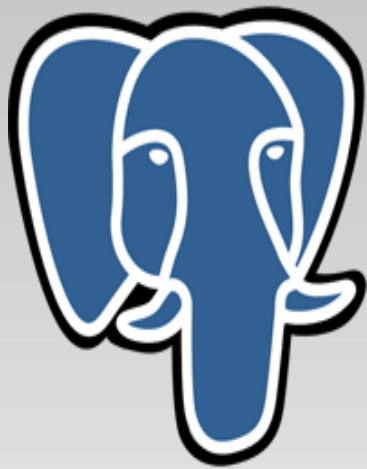




PostgreSQL

POSTGRES DATABASE ADMINISTRATION



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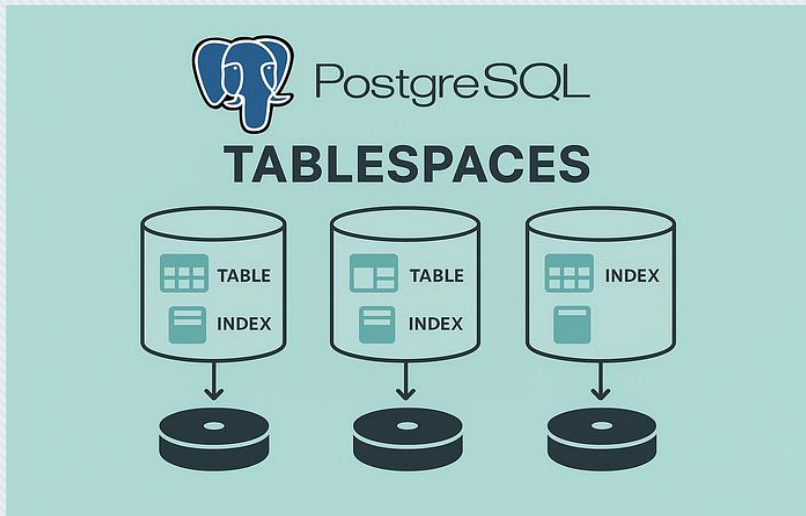
|SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS|

CHAPTER 3: POSTGRESQL PHYSICAL STRUCTURES – TABLESPACES.

About the Author:

I am **Stephen Njoroge**, a Senior Database Engineer with over 8+ years of expertise in designing, securing and optimizing high-performance database systems across the banking and fintech sectors. My core technical proficiency spans major databases: Oracle, PostgreSQL, MS SQL Server and MySQL-along with advanced scripting in SQL and Python for automation and data pipeline development. I architect and manage cloud and virtualization solutions on OCI, AWS, and Proxmox for onprem systems and implement comprehensive DevOps practices using monitoring tools like Prometheus and Grafana and IT service management via iTop. Further, I deliver advanced data analytics and visualization using Tableau, Power BI and Oracle BI. As a Certified Data Protection Officer, I integrate robust data protection, compliance and security measures including vulnerability assessment and role-based access control into every layer of the data infrastructure.

Tablespaces in PostgreSQL



Tablespaces are locations on disk where PostgreSQL stores data files (tables, indexes, etc.). They allow you to:

- Store data on different storage devices
- Manage disk space allocation
- Optimize performance by separating data types
- Control physical data placement

Tablespace Hierarchy

Operating System

- Tablespace 1: `pg_default` → `$PGDATA/base` (default location)
- Tablespace 2: `fast_ssd` → `/mnt/ssd/postgres_data`
- Tablespace 3: `slow_hdd` → `/mnt/hdd/postgres_archive`
- Tablespace 4: `indexes` → `/mnt/nvme/postgres_indexes`

Default Tablespaces

Every PostgreSQL cluster has two default tablespaces:

-- View default tablespaces

```
SELECT spcname, pg_tablespace_location(oid)
FROM pg_tablespace;
```

```
postgres=# SELECT spcname, pg_tablespace_location(oid)
FROM pg_tablespace;
 spcname | pg_tablespace_location 
-----|-----
 pg_default | 
 pg_global | 
 training_tbs | /var/lib/postgresql/training_tbs
(3 rows)

postgres=#
```

Creating Tablespaces

Prerequisites

- # 1. Create directory with proper permissions

```
sudo mkdir -p /mnt/ssd/postgres_data
sudo chown postgres:postgres /mnt/ssd/postgres_data
```
- # 2. Ensure PostgreSQL user has write access

```
sudo -u postgres touch /mnt/ssd/postgres_data/test.txt
```

```
root@998-986-02:/home/steve# sudo mkdir -p /mnt/ssd/postgres_data
root@998-986-02:/home/steve# sudo chown postgres:postgres /mnt/ssd/postgres_data
root@998-986-02:/home/steve# sudo -u postgres touch /mnt/ssd/postgres_data/test.txt
root@998-986-02:/home/steve# ls -lrt /mnt/ssd/postgres_data/test.txt
-rw-rw-r-- 1 postgres postgres 0 Jan 30 09:45 /mnt/ssd/postgres_data/test.txt
root@998-986-02:/home/steve#
```

Create Tablespace

-- Basic creation

```
CREATE TABLESPACE fast_ssd LOCATION '/mnt/ssd/postgres_data';
```

```
postgres=# CREATE TABLESPACE fast_ssd LOCATION '/mnt/ssd/postgres_data';
CREATE TABLESPACE
postgres=# \db
```

List of tablespaces		
Name	Owner	Location
fast_ssd	postgres	/mnt/ssd/postgres_data
pg_default	postgres	
pg_global	postgres	
training_tbs	postgres	/var/lib/postgresql/training_tbs

(4 rows)

```
postgres=#
```

-- With owner

--create tablespace

```
CREATE TABLESPACE archive_hdd
LOCATION '/mnt/hdd/postgres_archive';
```

```
postgres=# CREATE TABLESPACE archive_hdd
LOCATION '/mnt/hdd/postgres_archive';
CREATE TABLESPACE
postgres=#
```

--assign the owner.

```
ALTER TABLESPACE archive_hdd OWNER TO archive_user;
```

```
postgres=# ALTER TABLESPACE archive_hdd OWNER TO archive_user;
ALTER TABLESPACE
postgres=#
```

Managing Tablespaces

List Tablespaces

-- Basic List

```
\db
```

```
SELECT * FROM pg_tablespace;
```

```
postgres=# \db
```

List of tablespaces		
Name	Owner	Location
archive_hdd	archive_user	/mnt/hdd/postgres_archive
fast_ssd	postgres	/mnt/ssd/postgres_data
pg_default	postgres	
pg_global	postgres	
training_tbs	postgres	/var/lib/postgresql/training_tbs

```
(5 rows)
```

```
postgres=# SELECT * FROM pg_tablespace;
```

oid	spcname	spcowner	spcacl	spcoptions
1663	pg_default	10		
1664	pg_global	10		
16393	training_tbs	10		
26899	fast_ssd	10		
26903	archive_hdd	26900		

```
(5 rows)
```

```
postgres=#
```

-- Detailed information

```
SELECT
    ts.spcname as name,
    pg_tablespace_location(ts.oid) as location,
    pg_size_pretty(pg_tablespace_size(ts.spcname)) as size,
    ts.spcowner::regrole as owner,
    ts.spcacl as privileges,
    ts.spcoptions as options
FROM pg_tablespace ts
ORDER BY pg_tablespace_size(ts.spcname) DESC NULLS LAST;
```

```
postgres=# SELECT
    ts.spcname as name,
    pg_tablespace_location(ts.oid) as location,
    pg_size_pretty(pg_tablespace_size(ts.spcname)) as size,
    ts.spcowner::regrole as owner,
    ts.spcacl as privileges,
    ts.spcoptions as options
FROM pg_tablespace ts
ORDER BY pg_tablespace_size(ts.spcname) DESC NULLS LAST;
```

name	location	size	owner	privileges	options
pg_default		115 MB	postgres		
pg_global		468 kB	postgres		
training_tbs	/var/lib/postgresql/training_tbs	0 bytes	postgres		
fast_ssd	/mnt/ssd/postgres_data	0 bytes	postgres		
archive_hdd	/mnt/hdd/postgres_archive	0 bytes	archive_user		

```
(5 rows)
```

```
postgres=#
```

-- Show tablespace usage by database

```
SELECT
    d.datname as database,
    t.spcname as tablespace,
    count(*) as objects,
    pg_size_pretty(sum(pg_total_relation_size(c.oid))) as total_size
FROM pg_database d
CROSS JOIN pg_tablespace t
```

```

LEFT JOIN pg_class c ON c.reltablespace = t.oid
    OR (c.reltablespace = 0 AND d.dattablespace = t.oid)
WHERE d.datistemplate = false
GROUP BY d.datname, t.spcname
ORDER BY sum(pg_total_relation_size(c.oid)) DESC NULLS LAST;

```

database	tablespace	objects	total_size
myapp_dev	pg_default	370	11 MB
myapp_reporting	pg_default	370	11 MB
cloned_db	pg_default	370	11 MB
postgres	pg_default	370	11 MB
maintenance_db	pg_default	370	11 MB
newdb	pg_default	370	11 MB
myapp_prod	pg_default	370	11 MB
db2	pg_default	370	11 MB
training_db_restored	pg_default	370	11 MB
training_db	pg_default	370	11 MB
db1	pg_default	370	11 MB
training_db	pg_global	46	824 kB
db2	pg_global	46	824 kB
myapp_reporting	pg_global	46	824 kB
newdb	pg_global	46	824 kB
cloned_db	pg_global	46	824 kB
training_db_restored	pg_global	46	824 kB
postgres	pg_global	46	824 kB
maintenance_db	pg_global	46	824 kB
myapp_dev	pg_global	46	824 kB
myapp_prod	pg_global	46	824 kB
db1	pg_global	46	824 kB
maintenance_db	archive_hdd	1	
myapp_reporting	fast_ssd	1	
cloned_db	fast_ssd	1	
training_db_restored	fast_ssd	1	
db1	archive_hdd	1	
cloned_db	archive_hdd	1	
myapp_prod	archive_hdd	1	

Modify Tablespaces

```

-- Rename tablespace
ALTER TABLESPACE fast_ssd RENAME TO ssd_tablespace;

```

```

postgres=# ALTER TABLESPACE fast_ssd RENAME TO ssd_tablespace;
ALTER TABLESPACE
postgres=# \db

```

Name	Owner	Location
archive_hdd	archive_user	/mnt/hdd/postgres_archive
pg_default	postgres	
pg_global	postgres	
ssd_tablespace	postgres	/mnt/ssd/postgres_data
training_tbs	postgres	/var/lib/postgresql/training_tbs

```

(5 rows)

postgres=# |

```

```

-- Change owner
ALTER TABLESPACE ssd_tablespace OWNER TO prod_user;

```

```
postgres=# \db
```

List of tablespaces		
Name	Owner	Location
archive_hdd	archive_user	/mnt/hdd/postgres_archive
pg_default	postgres	
pg_global	postgres	
ssd_tablespace	postgres	/mnt/ssd/postgres_data
training_tbs	postgres	/var/lib/postgresql/training_tbs

(5 rows)

```
postgres=# ALTER TABLESPACE ssd_tablespace OWNER TO prod_user;
ALTER TABLESPACE
```

```
postgres=# \db
```

List of tablespaces		
Name	Owner	Location
archive_hdd	archive_user	/mnt/hdd/postgres_archive
pg_default	postgres	
pg_global	postgres	
ssd_tablespace	prod_user	/mnt/ssd/postgres_data
training_tbs	postgres	/var/lib/postgresql/training_tbs

(5 rows)

```
postgres=#
```

```
-- Set parameters
```

```
ALTER TABLESPACE ssd_tablespace SET (random_page_cost = 1.2);
```

```
postgres=# ALTER TABLESPACE ssd_tablespace SET (random_page_cost = 1.2);
```

```
ALTER TABLESPACE
```

```
postgres=# SELECT spcname, spcoptions
```

```
FROM pg_tablespace
```

```
WHERE spcname = 'ssd_tablespace';
```

spcname	spcoptions
ssd_tablespace	{random_page_cost=1.2}

(1 row)

```
postgres=# |
```

Drop Tablespaces

```
-- Check if tablespace is empty first
```

```
SELECT schemaname, tablename
```

```
FROM pg_tables
```

```
WHERE tablespace = 'archive_hdd';
```

```
-- Drop empty tablespace
```

```
DROP TABLESPACE archive_hdd;
```

```
postgres=# SELECT schemaname, tablename
FROM pg_tables
WHERE tablespace = 'archive_hdd';
 schemaname | tablename
-----+-----
(0 rows)
```

```
postgres=# DROP TABLESPACE archive_hdd;
DROP TABLESPACE
postgres=#
```

-- Move all objects from tablespace

```
ALTER TABLE orders SET TABLESPACE ts_ssd;
ALTER INDEX orders_pkey SET TABLESPACE ts_ssd;
```

```
postgres=# ALTER TABLE orders SET TABLESPACE ts_ssd;
ALTER TABLE
postgres=# ALTER INDEX orders_pkey SET TABLESPACE ts_ssd;
ALTER INDEX
postgres=# \db
```

Name	Owner	Location
pg_default	postgres	
pg_global	postgres	
ssd_tablespace	prod_user	/mnt/ssd/postgres_data
training_tbs	postgres	/var/lib/postgresql/training_tbs
ts_ssd	postgres	/mnt/ssd/pg_tablespace1

(5 rows)

```
postgres=# |
```

Using Tablespaces

Create Objects in Specific Tablespace

-- Database with custom tablespace

```
CREATE DATABASE mydb TABLESPACE ssd_tablespace;
```

```
postgres=# CREATE DATABASE mydb TABLESPACE ssd_tablespace;
CREATE DATABASE
postgres=#
```

-- Table in specific tablespace

```
CREATE TABLE tra_ining (
  id SERIAL PRIMARY KEY,
  name TEXT
) TABLESPACE ssd_tablespace;
```

```
postgres=# CREATE TABLE tra_ining (
  id SERIAL PRIMARY KEY,
  name TEXT
) TABLESPACE ssd_tablespace;
CREATE TABLE
postgres=#
```

-- Index in different tablespace

```
CREATE INDEX idx_users_name ON tra_ining (name)
TABLESPACE training_tbs;
```

```
postgres=# CREATE INDEX idx_users_name ON tra_ining (name)
TABLESPACE training_tbs;
CREATE INDEX
postgres=#
```

-- Partitioned table with tablespaces

```
CREATE TABLE sensor_data (
    id BIGSERIAL,
    sensor_id INT,
    reading FLOAT,
    recorded_at TIMESTAMP
) PARTITION BY RANGE (recorded_at)
TABLESPACE ssd_tablespace;
```

```
postgres=# CREATE TABLE sensor_data (
    id BIGSERIAL,
    sensor_id INT,
    reading FLOAT,
    recorded_at TIMESTAMP
) PARTITION BY RANGE (recorded_at)
TABLESPACE ssd_tablespace;
CREATE TABLE
postgres=# |
```

-- Each partition on different tablespace

```
CREATE TABLE sensor_data_2024_01 PARTITION OF sensor_data
FOR VALUES FROM ('2024-01-01') TO ('2024-02-01')
TABLESPACE ts_ssd;
```

```
postgres=# CREATE TABLE sensor_data_2024_01 PARTITION OF sensor_data
FOR VALUES FROM ('2024-01-01') TO ('2024-02-01')
TABLESPACE ts_ssd;
CREATE TABLE
postgres=# |
```

```
CREATE TABLE sensor_data_2024_02 PARTITION OF sensor_data
FOR VALUES FROM ('2024-02-01') TO ('2024-03-01')
TABLESPACE ts_ssd;
```

```
postgres=# CREATE TABLE sensor_data_2024_02 PARTITION OF sensor_data
FOR VALUES FROM ('2024-02-01') TO ('2024-03-01')
TABLESPACE ts_ssd;
CREATE TABLE
postgres=#
```

Move Objects Between Tablespaces

```
-- Move table to different tablespace
ALTER TABLE users SET TABLESPACE ts_ssd;
```

```
postgres=# ALTER TABLE users SET TABLESPACE ts_ssd;
ALTER TABLE
postgres=#
```

-- Move index

```
ALTER INDEX orders_pkey SET TABLESPACE ts_ssd;
```

```
postgres=# ALTER INDEX orders_pkey SET TABLESPACE ts_ssd;
ALTER INDEX
postgres=#
```

-- Move all objects from one tablespace to another

```
DO $$
DECLARE
    r RECORD;
BEGIN
    FOR r IN (
        SELECT schemaname, tablename
        FROM pg_tables
        WHERE tablespace = 'ts_ssd'
    ) LOOP
        EXECUTE format('ALTER TABLE %I.%I SET TABLESPACE ssd_tablespace',
            r.schemaname, r.tablename);
    END LOOP;

    FOR r IN (
        SELECT schemaname, indexname
        FROM pg_indexes
        WHERE tablespace = 'ts_ssd'
    ) LOOP
        EXECUTE format('ALTER INDEX %I.%I SET TABLESPACE ssd_tablespace',
            r.schemaname, r.indexname);
    END LOOP;
END $$;
```

```
SELECT
    schemaname AS schema,
    tablename AS table,
    COALESCE(tablespace, 'default') AS tablespace
FROM pg_tables
ORDER BY schemaname, tablename;
```

public	orders	ssd_tablespace
public	payments	default
public	products	default
public	sensor_data	ssd_tablespace
public	sensor_data_2024_01	ssd_tablespace
public	sensor_data_2024_02	ssd_tablespace
public	tra_ining	ssd_tablespace
public	users	ssd_tablespace

(77 rows)

Tablespace Parameters

Cost-Based Parameters

```
-- Set cost parameters for tablespace
ALTER TABLESPACE fast_ssd SET (
    random_page_cost = 1.0,    -- Lower for SSDs
    seq_page_cost = 1.0       -- Lower for fast storage
);
```

```
postgres=# ALTER TABLESPACE fast_ssd SET (
    random_page_cost = 1.0,    -- Lower for SSDs
    seq_page_cost = 1.0       -- Lower for fast storage
);
ALTER TABLESPACE
postgres=#
```

```
ALTER TABLESPACE slow_hdd SET (
    random_page_cost = 4.0,    -- Higher for HDDs
    seq_page_cost = 1.0
);
```

```
postgres=# ALTER TABLESPACE slow_hdd SET (
    random_page_cost = 4.0,    -- Higher for HDDs
    seq_page_cost = 1.0
);
ALTER TABLESPACE
postgres=# |
```

```
-- View current parameters
SELECT
    spcname,
    spcoptions
FROM pg_tablespace
WHERE spcoptions IS NOT NULL;
```

```
postgres=# SELECT
    spcname,
    spcoptions
FROM pg_tablespace
WHERE spcoptions IS NOT NULL;
 spcname | spcoptions
-----|-----
ssd_tablespace | {random_page_cost=1.2}
fast_ssd | {random_page_cost=1.0,seq_page_cost=1.0}
slow_hdd | {random_page_cost=4.0,seq_page_cost=1.0}
(3 rows)

postgres=# |
```

Monitoring Tablespaces

Disk Usage

-- Size of each tablespace

```
SELECT
    spcname,
    pg_size_pretty(pg_tablespace_size(spcname)) as size,
    pg_tablespace_location(oid) as location
FROM pg_tablespace;
```

```
postgres=# SELECT
    spcname,
    pg_size_pretty(pg_tablespace_size(spcname)) as size,
    pg_tablespace_location(oid) as location
FROM pg_tablespace;
 spcname | size | location
-----|-----|-----
pg_default | 115 MB |
pg_global | 468 kB |
training_tbs | 12 kB | /var/lib/postgresql/training_tbs
ssd_tablespace | 7649 kB | /mnt/ssd/postgres_data
ts_ssd | 4096 bytes | /mnt/ssd/pg_tablespace1
fast_ssd | 0 bytes | /mnt/fast_ssd/postgres
slow_hdd | 0 bytes | /mnt/hdd/postgres_slow
(7 rows)

postgres=# |
```

-- Detailed space usage

```
SELECT
    ts.spcname as tablespace,
    n.nspname as schema,
    c.relname as object,
    CASE c.relkind
        WHEN 'r' THEN 'table'
        WHEN 'i' THEN 'index'
        WHEN 'S' THEN 'sequence'
        ELSE c.relkind::text
    END as type,
    pg_size_pretty(pg_total_relation_size(c.oid)) as size
FROM pg_class c
JOIN pg_tablespace ts ON ts.oid = c.reltablespace
JOIN pg_namespace n ON n.oid = c.relnamespace
WHERE c.reltablespace > 0
ORDER BY pg_total_relation_size(c.oid) DESC;
```

tablespace	schema	object	type	size
pg_global	pg_catalog	pg_shdepend	table	80 kB
pg_global	pg_catalog	pg_auth_members	table	72 kB
ssd_tablespace	public	users	table	64 kB
pg_global	pg_catalog	pg_database	table	48 kB
pg_global	pg_catalog	pg_tablespace	table	48 kB
pg_global	pg_catalog	pg_authid	table	40 kB
pg_global	pg_catalog	pg_db_role_setting	table	32 kB
pg_global	pg_catalog	pg_shdescription	table	32 kB
ssd_tablespace	public	orders	table	32 kB
ssd_tablespace	public	tra_ining	table	24 kB
pg_global	pg_catalog	pg_parameter_acl	table	24 kB
pg_global	pg_catalog	pg_subscription	table	24 kB
pg_global	pg_catalog	pg_auth_members_member_role_index	index	16 kB
ssd_tablespace	public	orders_pkey	index	16 kB
pg_global	pg_catalog	pg_database_datname_index	index	16 kB
pg_global	pg_catalog	pg_database_oid_index	index	16 kB
pg_global	pg_catalog	pg_db_role_setting_databaseid_rol_index	index	16 kB
pg_global	pg_catalog	pg_tablespace_oid_index	index	16 kB
pg_global	pg_catalog	pg_tablespace_spcname_index	index	16 kB
pg_global	pg_catalog	pg_authid_rolname_index	index	16 kB
pg_global	pg_catalog	pg_authid_oid_index	index	16 kB
pg_global	pg_catalog	pg_auth_members_oid_index	index	16 kB
pg_global	pg_catalog	pg_auth_members_role_member_index	index	16 kB
pg_global	pg_catalog	pg_auth_members_grantor_index	index	16 kB
pg_global	pg_catalog	pg_shdepend_depender_index	index	16 kB
pg_global	pg_catalog	pg_shdepend_reference_index	index	16 kB

-- Top 10 Largest objects in tablespace

```
SELECT
    n.nspname as schema,
    c.relname as object,
    pg_size_pretty(pg_total_relation_size(c.oid)) as size,
    pg_size_pretty(pg_relation_size(c.oid)) as table_size,
    pg_size_pretty(pg_indexes_size(c.oid)) as index_size
FROM pg_class c
JOIN pg_namespace n ON n.oid = c.relnamespace
WHERE c.reltablespace = (
    SELECT oid FROM pg_tablespace WHERE spcname = 'ssd_tablespace'
)
ORDER BY pg_total_relation_size(c.oid) DESC
LIMIT 10;
```

schema	object	size	table_size	index_size
public	users	64 kB	8192 bytes	48 kB
public	orders	32 kB	8192 bytes	16 kB
public	tra_ining	24 kB	0 bytes	16 kB
public	orders_pkey	16 kB	16 kB	0 bytes
pg_toast	pg_toast_26905_index	8192 bytes	8192 bytes	0 bytes
pg_toast	pg_toast_26935_index	8192 bytes	8192 bytes	0 bytes
pg_toast	pg_toast_26993	8192 bytes	0 bytes	8192 bytes
pg_toast	pg_toast_26993_index	8192 bytes	8192 bytes	0 bytes
pg_toast	pg_toast_26935	8192 bytes	0 bytes	8192 bytes
pg_toast	pg_toast_26905	8192 bytes	0 bytes	8192 bytes

(10 rows)

postgres=#

Disk Free Space

```
# Check disk space for tablespace locations
df -h /mnt/ssd /mnt/hdd /mnt/nvme
```

```

Filesystem      Size  Used Avail Use% Mounted on
/dev/sdc        1007G  3.3G  953G   1% /
/dev/sdc        1007G  3.3G  953G   1% /

```

```

# Check inode usage
df -i /mnt/ssd

```

```

Filesystem      Inodes IUsed   IFree IUse% Mounted on
/dev/sdc        67108864 71772 67037092   1% /

```

Tablespace Management Commands

```

# Create tablespace directory and set permissions
sudo mkdir -p /mnt/tablespaces/{ssd,hdd,nvme}

```

```

root@998-986-02:/home/steve# sudo mkdir -p /mnt/tablespaces/{ssd,hdd,nvme}
root@998-986-02:/home/steve# ls -lrt /mnt/tablespaces
total 12
drwxr-xr-x 2 root root 4096 Jan 30 13:01 ssd
drwxr-xr-x 2 root root 4096 Jan 30 13:01 hdd
drwxr-xr-x 2 root root 4096 Jan 30 13:01 nvme

```

```

sudo chown -R postgres:postgres /mnt/tablespaces/

```

```

sudo chmod 700 /mnt/tablespaces/*

```

```

root@998-986-02:/home/steve# sudo chown -R postgres:postgres /mnt/tablespaces/
root@998-986-02:/home/steve# sudo chmod 700 /mnt/tablespaces/*
root@998-986-02:/home/steve#

```

```

# List tablespaces from command line

```

```

postgres=# \db+

```

Name	Owner	Location	List of tablespaces		Options	Size	Description
			Access privileges				
fast_ssd	postgres	/mnt/fast_ssd/postgres			{random_page_cost=1.0,seq_page_cost=1.0}	0 bytes	
pg_default	postgres					115 MB	
pg_global	postgres					468 kB	
slow_hdd	postgres	/mnt/hdd/postgres_slow			{random_page_cost=4.0,seq_page_cost=1.0}	0 bytes	
ssd_tablespace	prod_user	/mnt/ssd/postgres_data			{random_page_cost=1.2}	7649 kB	
training_tbs	postgres	/var/lib/postgresql/training_tbs				12 kB	
ts_ssd	postgres	/mnt/ssd/pg_tablespace1				4096 bytes	
(7 rows)							

Tablespace Strategies

Data Lifecycle Management

```

-- Hot data on fast storage
CREATE TABLESPACE hot_data LOCATION '/mnt/nvme/hot_data';

-- Warm data on SSD

```

```
CREATE TABLESPACE warm_data LOCATION '/mnt/ssd/warm_data';

-- Cold data on HDD
CREATE TABLESPACE cold_data LOCATION '/mnt/hdd/cold_data';
```

```
postgres=# CREATE TABLESPACE hot_data LOCATION '/mnt/nvme/hot_data';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE warm_data LOCATION '/mnt/ssd/warm_data';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE cold_data LOCATION '/mnt/hdd/cold_data';
CREATE TABLESPACE
postgres=#
```

Index Separation

```
-- Tables on SSDs
CREATE TABLESPACE data_ssd LOCATION '/mnt/ssd/data';

-- Indexes on NVMe
CREATE TABLESPACE indexes_nvme LOCATION '/mnt/nvme/indexes';
```

```
postgres=# CREATE TABLESPACE data_ssd LOCATION '/mnt/ssd/data';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE indexes_nvme LOCATION '/mnt/nvme/indexes';
ERROR: directory "/mnt/nvme/indexes" does not exist
postgres=# CREATE TABLESPACE indexes_nvme LOCATION '/mnt/nvme/indexes';
CREATE TABLESPACE
postgres=#
```

-- Create table with indexes on different tablespace

```
CREATE TABLE orders_partitioned (
    id BIGSERIAL PRIMARY KEY,
    customer_id INT,
    amount DECIMAL(10,2),
    order_date DATE
) TABLESPACE data_ssd;
```

```
postgres=# CREATE TABLE orders_partitioned (
    id BIGSERIAL PRIMARY KEY,
    customer_id INT,
    amount DECIMAL(10,2),
    order_date DATE
) TABLESPACE data_ssd;
CREATE TABLE
postgres=# |
```

-- Create index on fast storage

```
CREATE INDEX idx_orders_partitioned_customer
ON orders_partitioned(customer_id)
TABLESPACE indexes_nvme;
```

```
postgres=# CREATE INDEX idx_orders_partitioned_customer
ON orders_partitioned(customer_id)
TABLESPACE indexes_nvme;
CREATE INDEX
postgres=# |
```

```
CREATE INDEX idx_orders_date ON orders(order_date)
```

```
TABLESPACE indexes_nvme;
```

```
CREATE INDEX idx_orders_date ON orders(order_date)
TABLESPACE indexes_nvme;
CREATE INDEX
postgres=#
```

Strategy 3: Multi-tenant Database

-- Different tenants on different storage

```
CREATE TABLESPACE tenant_a LOCATION '/mnt/ssd/tenant_a';
CREATE TABLESPACE tenant_b LOCATION '/mnt/ssd/tenant_b';
```

```
postgres=# CREATE TABLESPACE tenant_a LOCATION '/mnt/ssd/tenant_a';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE tenant_b LOCATION '/mnt/ssd/tenant_b';
CREATE TABLESPACE
```

```
CREATE TABLESPACE tenant_c LOCATION '/mnt/hdd/tenant_c';
```

```
postgres=# CREATE TABLESPACE tenant_c LOCATION '/mnt/hdd/tenant_c';
CREATE TABLESPACE
postgres=#
```

-- Each tenant's schema in different tablespace

```
CREATE SCHEMA tenant_a;
CREATE TABLE tenant_a.users (...);
ALTER TABLE tenant_a.users SET TABLESPACE tenant_a;
```

```
postgres=# CREATE TABLE tenant_a.users (
    id BIGSERIAL PRIMARY KEY,
    username TEXT NOT NULL,
    email TEXT,
    created_at TIMESTAMP DEFAULT now()
);
CREATE TABLE
postgres=# ALTER TABLE tenant_a.users SET TABLESPACE tenant_a;
ALTER TABLE
postgres=#
```

```
CREATE SCHEMA tenant_b;
CREATE TABLE tenant_b.users (...);
ALTER TABLE tenant_b.users SET TABLESPACE tenant_b;
```

```
postgres=# CREATE TABLE tenant_b.users (
    id BIGSERIAL PRIMARY KEY,
    username TEXT NOT NULL,
    email TEXT,
    created_at TIMESTAMP DEFAULT now()
);
CREATE TABLE
postgres=# ALTER TABLE tenant_b.users
SET TABLESPACE tenant_b;
ALTER TABLE
postgres=#
```

Backup & Recovery

Tablespace-Aware Backup

Backup with tablespaces

```
pg_dump -Fc --create --verbose --file=backup.dump mydb
```

```
postgres@998-986-02:/home/steve$ pg_dump -Fc --create --verbose --file=backup.dump mydb
pg_dump: executing SELECT pg_catalog.set_config('search_path', '', false);
pg_dump: last built-in OID is 16383
pg_dump: reading extensions
pg_dump: identifying extension members
pg_dump: reading schemas
pg_dump: reading user-defined tables
pg_dump: reading user-defined functions
pg_dump: reading user-defined types
pg_dump: reading procedural languages
pg_dump: reading user-defined aggregate functions
pg_dump: reading user-defined operators
pg_dump: reading user-defined access methods
pg_dump: reading user-defined operator classes
pg_dump: reading user-defined operator families
pg_dump: reading user-defined text search parsers
pg_dump: reading user-defined text search templates
pg_dump: reading user-defined text search dictionaries
pg_dump: reading user-defined text search configurations
pg_dump: reading user-defined foreign-data wrappers
```

Backup all databases with tablespaces

```
pg_dumpall --globals-only > globals.sql
```

```
postgres@998-986-02:/home/steve$ pg_dumpall --globals-only > globals.sql
postgres@998-986-02:/home/steve$
```

Changing Default Tablespace in PostgreSQL

Understanding Default Tablespaces

By default, PostgreSQL uses:

- `pg_default` tablespace for user data (located in `$PGDATA/base/`)
- `pg_global` tablespace for system catalogs

Check Current Default Tablespace

-- Check current default tablespace for the database

```
SELECT datname, dattablespace, ts.spcname
FROM pg_database d
JOIN pg_tablespace ts ON d.dattablespace = ts.oid
```

```
WHERE datname = current_database();
```

```
postgres=# SELECT datname, dattablespace, ts.spcname
FROM pg_database d
JOIN pg_tablespace ts ON d.dattablespace = ts.oid
WHERE datname = current_database();
 datname | dattablespace | spcname
-----+-----+-----
 postgres |          1663 | pg_default
(1 row)

postgres=#
```

```
-- Check database-wide default
SELECT datname, dattablespace
FROM pg_database
WHERE datname = 'mydb';
```

```
postgres=# SELECT datname, dattablespace
FROM pg_database
WHERE datname = 'mydb';
 datname | dattablespace
-----+-----
 mydb    |          26899
(1 row)

postgres=# |
```

Change Default Tablespace at Different Levels

Change for a SPECIFIC DATABASE

```
-- 1. Create new tablespace first
CREATE TABLESPACE fast_ssd2 LOCATION '/mnt/ssd/postgres_data2';
```

```
postgres=# CREATE TABLESPACE fast_ssd2 LOCATION '/mnt/ssd/postgres_data2';
CREATE TABLESPACE
postgres=# |
```

```
-- 2. Change database's default tablespace
ALTER DATABASE mydb SET TABLESPACE fast_ssd2;
```

```
postgres=# ALTER DATABASE mydb SET TABLESPACE fast_ssd2;
ALTER DATABASE
postgres=# |
```

Change for CURRENT SESSION Only

```
-- Set default for current session (temporary)
SET default_tablespace = 'fast_ssd2';
```

```
postgres=# SET default_tablespace = 'fast_ssd2';
SET
postgres=# |
```

```
-- Verify
SHOW default_tablespace;
```

```
postgres=# SHOW default_tablespace;
 default_tablespace
-----
    fast_ssd2
(1 row)

postgres=# |
```

```
-- All new tables/indexes in this session will use this tablespace
CREATE TABLE temp_session_table (id SERIAL);
```

```
postgres=# CREATE TABLE temp_session_table (id SERIAL);
CREATE TABLE
postgres=# |
```